# Planetary Rover Distributed Sensing and Cooperation


**Abstract**

This report outlines a planning architecture for a multi-rover planetary exploration mission. The system utilizes agents with asymmetric capabilities: highly mobile Navigator rovers for exploration and sample retrieval, and a heavy Spectrometer rover for in-situ analysis. The domain is initially modeled using classical PDDL to enforce physical and data-driven cooperation, and subsequently extended to PDDL+ to model continuous real-world constraints such as data latency, signal degradation, and temporal feasibility.


## 1 Introduction

Autonomous multi-agent systems are critical for planetary surface exploration. To optimize payload capacity and energy constraints, modern missions often deploy rovers with heterogeneous capabilities. This project explores the planning complexities of such asymmetric systems. By forcing cooperation between sensing-deprived mobile agents and a stationary analysis agent, the model evaluates the trade-offs between mobility and information gathering using symbolic planning solvers.

## 2 Methodology & Results

### 2.1 Part 1: Basic PDDL Model (Discrete Logic)

The foundation is a graph-based spatial model encoded in PDDL. The environment is abstracted into a waypoint graph using a `(connected ?l1 ?l2)` predicate. Within the context of symbolic planning, this creates a "fictional interface" that abstracts continuous geometric constraints to keep the symbolic search tractable.

The agents are divided strictly by their capabilities: **Navigator Rovers** can move and grip light objects but cannot analyze them, while the **Spectrometer Rover** contains the sole analysis hardware. Analysis is restricted via `:disjunctive-preconditions`; the Spectrometer can only analyze a sample if it is at the sample's location AND either the sample was physically delivered OR its coordinates were transmitted.

- **Physical Cooperation:** For light, movable samples, the Navigator uses `collect` and `deliver` actions to physically drop the sample at the Spectrometer's location.
- **Data Cooperation:** For heavy, immovable samples, the Navigator logs the coordinates via `save_samplelocation` and passes a boolean flag via the `transmit_receive_comm` action.

Using the ENHSP planner, a dual-sample mission successfully generated a 14-step plan demonstrating both physical delivery and data transfer.

### 2.2 Part 2: PDDL+ Model (Continuous Time)

To model realistic communication constraints, the domain was upgraded to PDDL+, shifting to a hybrid dynamical system using continuous time (`#t`).

- **Continuous Processes:** A `start-transmit` action triggers the `data-latency` process, which autonomously increases the `(data-transferred)` numeric fluent based on a defined `transfer-rate` over time.
- **Autonomous Events:** The `transfer-complete` event fires automatically when the data threshold is met. Conversely, if the Navigator exceeds the `comm-threshold` distance during transmission, a `data-loss` event abruptly fires, resetting data to 0 to simulate RF signal degradation.

The planner successfully handles these temporal constraints. For a 10.0-unit data transfer at 2.0 units/second, the ENHSP output explicitly injects a `-----waiting---- [5.0]` block, mathematically forcing the robots to remain stationary to prevent the `data-loss` event from ruining the plan.

## 3 Discussion & Analysis

The modeled system highlights the core trade-off between mobility and sensing. Centralizing analysis hardware creates a high-value asset, reducing overall mission weight but creating a logistical bottleneck. The symbolic plans prove it is far more energy-efficient for lightweight Navigators to explore wide perimeters and transmit data than to move the heavy Spectrometer continuously.

Furthermore, upgrading the domain to PDDL+ highlights the limitations of classical PDDL. While standard PDDL abstracts communication as an instantaneous logic flip, PDDL+ treats it as a critical, fragile physical resource, forcing the planner to accurately reason about time, bandwidth, and spatial limits.

## 4 Future Work & Scalability

- **Decentralized Planning:** Scaling to a massive N-rover swarm introduces the state explosion problem, requiring a shift from a global solver to localized, decentralized planning architectures.
- **Bridging the Task-Motion Gap:** The current model uses idealized geometric assumptions. Future architectures will use semantic attachments to call external continuous solvers (e.g., RRT) to validate collision-free trajectories in the Configuration Space during the search phase.
- **Reactive Control:** Local execution will integrate Morse Potential Functions, treating rovers as repelling/attracting forces to dynamically avoid physical collisions without constantly triggering symbolic replanning.

## 5 References


1. *Autonomous mobile surveying for science rovers using in situ distributed remote sensing*
2. *Cognition Distributed Data Processing System for Lunar Activities*
3. *Low cost multi-agent systems for planetary surface exploration*
4. *Motion Trajectories for Wide-Area Surveying with a Rover-Based Distributed Spectrometer*